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The Effects of a Normative Intervention on Group Decision-Making Performance

DOUGLAS McGREGOR (1957), in commenting upon the disadvantages of confusion, frustration, and time-loss which are frequently ascribed to group activities, has observed that such need not be the case. Indeed, he has suggested that it is possible to *create* decision-making groups which have few if any of these characteristics and which function efficiently and produce qualitatively superior decisions in the process. Essentially, McGregor's comments focus on one of the abiding concerns in social psychology – that of group effectiveness – and address particularly the discrepancy between commonly encountered group phenomena and the performance potentials which research in group dynamics has revealed. It is McGregor's contention that there is nothing inherent in groups qua groups which will account for their performance, either good or bad, and that the key to group effectiveness may best be stated in terms of the particular arrangement of systemic elements which characterizes a given group. Moreover, he maintains that group systems, because of their dynamic properties, are amenable to influence to such an extent that dysfunctional practices may be remedied and group performance upgraded through interventions aimed at the elements of process. Implicit in this position is the notion that active intervention into ongoing groups in the interest of more effective functioning is a valid undertaking for behavioral scientists.

In many respects McGregor has pinpointed an issue of both practical and theoretical significance. While placing new emphasis on the art of the possible where group performance is concerned, his remarks also serve to emphasize the need for basic research on those elements of process which are amenable to influence and the efficacy of various intervention strategies. Obviously, the success of any intervention attempt will be determined by both its feasibility and theoretical validity. Unfortunately, little research has been done on this facet of group dynamics, and would-be interventionists are generally lacking in information about empirically validated interventions. Aside from the work of Maier and his colleagues (Maier, 1950; Maier & Solem, 1952; and Maier & Hoffman, 1960) on the effects of a discussion leader promoting a developmental approach to problem solving while eliciting minority opinions and a recent study by Hall & Williams (1970) of the efficacy of a laboratory training intervention, few articles have appeared in the literature which deal with the modification of group decision-making processes. As a result, the literature is devoted primarily to accounts of how groups tend to behave under normal (or laboratory) conditions and enumerations of those factors which are associated with group effectiveness, with little data on the movement from one level of performance to the next higher plane.

Needed, therefore, are basic studies with applied implications. Investigations of those systemic relationships which are amenable to alteration, the specific interventions which can effect given changes, and the consequences for quality and commitment of the ensuing processes would seem to be in order if the synergistic promise suggested by McGregor is to be realized on a workaday basis. The present study constitutes such an investigation, and it is concerned with the efficacy of an intervention directed at the normative system which governs the manner in which a group will approach its decision task, utilize its member resources, and resolve its substantive conflicts. As such, the study will test the assumption that much of a group's performance – whether effective or ineffective – is a function of both the attitudes of group members toward group potential and their shared perceptions of what constitutes appropriate member behavior. The objective of the intervention employed will be to legitimize a number of behaviors commonly viewed as deviant and disruptive acts in groups and, at the same time, to set the procedural goal of groups at a higher performance level than most member attitudes would typically allow or support.

RATIONALE FOR THE INTERVENTION

In their study of the effects of laboratory training in group dynamics on group decision-making performance, Hall & Williams (1970) had an opportunity for comparing directly the activities of procedurally wise groups with those of less sophisticated and more typical control groups. Although the procedural differences detected were only alluded to briefly in the article, the observations and inferences the authors were able to make provided the rationale for the present study.

The performance of control groups was consistently below that of trained groups on measures of decision quality, utilization of resources, creativity, and the realization of synergistic potential; many of the differences could be traced to procedural factors, and it was reasoned that the orientation of untrained groups toward decision tasks was such that they were both unwilling and unable to optimize. Two procedural factors particularly seemed to differentiate the functioning of control groups from the performance of trained groups. The first factor, an inferred dynamic, was what may be thought of as a strain toward convergence; untrained group members, through comments and actions, exhibited an apparently common need to coalesce rapidly so that they could reach a decision and discharge the responsibility with which they were charged. It was, as one subject put it, as if the groups were more concerned with a commitment to reaching a decision than they were with their commitment to the decisions reached. Convergence seemed to be valued as a precondition to decision making, rather than as a natural by-product of such activities; and the strain toward convergence appeared to introduce a sense of urgency in groups which, in turn, had implications for the decision techniques employed and the manner in which opinion differences were handled. Finally, the strain seemed to operate in a typical goal-gradient fashion; beyond a certain minimal level, the closer groups came to achieving convergence, the harder they were observed to press for closure. The fact that the convergence strain was often observed to preempt the importance of both the decision task and members' emotional reactions was taken as a significant point of departure between untrained and trained group processes.

The second factor, which stems directly from the convergence strain phenomenon, concerned the resolution of conflicts. Given the goal-gradient nature of the covergence strain, it not only appeared to increase in strength as groups approached their standard of agreement, but it became more susceptible to frustration simultaneously. The range of opinion differences on a decision task constitutes the most obvious limitation on convergence in groups, and the reactions to such differences which members experience will determine not only how the differences are reconciled, but the degree of convergence actually achieved. In the case of untrained groups, opinion differences served to frustrate group members; the degree of frustration elicted seemed to be proportional to the number of members who had converged at that point. The effect of this factor on group process was essentially one of truncation. The nearer a group came to achieving convergence, the less tolerant it was of opinion differences, and the more likely it was to suppress differences or – in what amounted to leaving the field – capitulate to a minority so that the group could continue to move. Such practices were later found to have implications for both decision quality and member commitment, although the group process employed was in the best tradition of socially acceptable behaviors.

By contrast, trained groups were observed to avoid premature convergence and to actively promote divergence during the early stages of deliberation in an apparent attempt to gain expression of individual opinions and rationales. In effect, they promoted conflict which, in turn, they treated as symptomatic of incomplete sharing of information among members. Not only were opinion differences encouraged and tolerated, but they were resolved on the basis of a consensus at both substantive and emotional levels of discourse. Moreover, opinion differences were viewed as natural resultants of multiperson situations, and, as a result, generated little frustration among members. Convergence became less of a concern than the manner in which it was to be achieved. It is noteworthy that, despite the different orientations of trained and untrained groups in the study, they did not differ in the amount of time it took them to reach their final decisions. The major impact was on group effectiveness as revealed in decision quality.

The authors concluded that the observed differences in group process were essentially due to different normative orientations toward group activities, the appropriateness of certain behaviors in groups, and the criterion of convergence. The training intervention employed was directed at the *attitudinal* supports for such normative orientations, in an attempt to change attitudes toward group action and, as a result, the norms of process which would prevail. It was reasoned that it might also be possible to elicit more effective group action, however, without going through the rather time-consuming process of attitude change. If, in fact, the results obtained by Hall & Williams (1970) were due in part to the operation of implicit group norms which differed for untrained as opposed to trained groups, an intervention aimed specifically at the normative system which was to prevail during group work should also result in the type of performance increments noted among trained groups. Therefore, it was hypothesized that a normative statement which would break the strain toward convergence and require a consensual resolution of conflicts – while specifying a number of confronting and obstructive behaviors as legitimate and required – would elicit and sustain a group process which, irrespective of member attitudes, would allow untrained groups to function more effectively than they normally would under the normative system which they themselves would bring to the enterprise. This, then, was the approach adopted for the present study, and a normative intervention was designed for use with groups unfamiliar with group dynamics principles.

HYPOTHESES

It was assumed that the status of the experimenters, given the context within which the study was conducted, would serve to legitimize sufficiently the normative intervention so that subjects would make an honest effort to abide by its provisions. To the extent that subjects were in fact both willing and able to adhere to the groundrules set forth in the normative statement, it was hypothesized that:

Hypothesis 1: Instructed groups would produce decisions which were qualitatively superior to those produced by uninstructed control groups.

Hypothesis 2: Instructed groups would make more effective use of their available member resources than uninstructed control groups would.

Hypothesis 3: Instructed groups would achieve significantly more creativity, as revealed in emergent solutions, than would uninstructed groups.

Hypothesis 4: Instructed groups would functional synergistically, *i.e.*, perform at a higher level than their most skilled members, to a greater extent than uninstructed groups would.

EXPERIMENTAL DESIGN

SETTING AND SUBJECTS

The study was conducted in connection with six management seminars which were held for the middle and upper management personnel of several small business organizations. The 148 seminar participants served as subjects for the study, and at the beginning of the programs these individuals were assigned to 32 discussion groups of from four to six members each. Groups within each seminar were equated insofar as possible with respect to the ages and levels of managerial responsibility of members. Prior to the experimental period, subjects had been exposed to various presentations on organizational theory, delegation, labor relations, and the like and had spent an average of six hours in small discussion group activities. The groups studied, therefore, may be thought of as constituting quasi-established entities; that is, while they were beyond the ad hoc stage in their development, they lacked the lengthy history of interaction necessary for fully established groups.

TASK AND PROCEDURE

Toward the end of each seminar, participants engaged in a group decision-making exercise as a scheduled part of their seminar experience. Participants remained in their original discussion groups for purposes of the exercise, and groups were randomly assigned to the control and experimental conditions. In two seminars there were too few groups to split them into control and experimental groupings, so the first seminar was treated as a control for the subsequent experimental seminar. In all, 16 groups served as controls and 16 groups were employed in the experimental condition.

The decision task. The decision task employed was the NASA Moon Survival Problem (Hall, 1963). This task requires subjects to rank in order of their importance for survival some 15 items of equipment. Thus, the total decision product for both individuals and groups is composed of 15 interdependent judgments. The NASA Moon Survival Problem concerns the plight of the crew of an ill-fated space flight; background information supplied to subjects indicates that they are to think of themselves as crew members. The story line indicates that their spaceship was originally scheduled to rendezvous with a mother ship on the lighted surface of the moon; due to mechanical difficulties, however, they have been forced to crashland some 200 miles from the rendezvous point. It is further indicated that with the exception of 15 items all equipment was damaged beyond use during the crashlanding, and, since survival depends upon the crew reaching the mother ship, the available equipment must be evaluated with respect to its importance for insuring survival during the crew's 200 mile cross-country trek. Subjects are asked to rank in order the 15 items in terms of their relative value and utility for survival.

A number of frames of reference may be employed in ranking the items, and it is particularly necessary for one to break his earth-bound set in order to perform well on the task. An expert answer for the task has been obtained from the Crew Equipment Research Section of the NASA Manned Spacecraft Center at Houston, Texas[1] and performance on the task can therefore be evaluated on the basis of an objectively correct criterion. The task has been found to generate extremely high levels of ego-involvement on the part of subjects, and decision adequacies have been found to be sensitive to a number of substantive and procedural contributions. Thus, the decision task employed affords a reasonable analogue of commonly encountered multi-stage decision-making situations.

Procedures for establishing normative conditions. All groups in the study received identical presentations of background information and task objectives. Subjects were first asked to rank in order the 15 items under alone conditions; each person was asked to fill out his rank order in duplicate so that one copy could be handed in to the experimenter for scoring while the other was retained by the subject as a guide for the group discussion which followed. When all subjects had completed both copies of their rankings and the experimenter's copies had been collected, subjects were asked to join with their discussion groups for purposes of arriving at a group decision on the NASA task. No time limits were imposed and groups were told that they were to report back to the experimenter with their group decisions whenever they were satisfied with the quality of their product.

At this point, half the groups were sent to their respective group meeting rooms without further instructions. These 16 groups were left to their own devices in

1. Copies of the NASA Moon Survival Problem and the objectively correct solution provided by NASA, appear as an Appendix to this article. The senior author is indebted to Mr. Matthew Radnofsky and Dr. Robert B. Voas of the National Aeronautics and Space Administration for their help in obtaining expert judgements on the task. Mr. Radnofsky, at the time the task was designed, was Chief of the Crew Equipment Research Department of NASA's Manned Spacecraft Center in Houston, Texas. Five of his personnel consolidated their individual rankings in arriving at the final expert criterion. The interjudge reliability prior to consolidation, as determined by the coefficient of concordance, was .82.

reaching a group decision, and they constitute, therefore, the control groups for the study. The remaining 16 groups received special instructions regarding the manner in which they were to work on their decision task; that is, the 16 experimental groups received 'normative' process instructions designed to orient them toward a 'working-through' approach to the handling of conflicts in the group. These instructions were presented in written form and elaborated upon verbally before subjects began their group work. The instructions given to the experimental groups for purposes of defining and legitimizing new norms of procedure are summarized below:

> *Instructions:* This is an exercise in group decision making. Your group is to employ the method of *group consensus* in reaching its decision. This means that the ranking for each of the 15 survival items must be agreed upon by each group member before it becomes a part of the group decision. Consensus is difficult to reach. Therefore, not every ranking will meet with everyone's *complete* approval. Unanimity, however, is not a goal (although it may be achieved unintentionally), and it is not necessary that every person be as satisfied as he might be, for example, if he had complete control over what the group decides. What should be stressed is the individual's ability to *accept* a given ranking on the basis of logic – whatever his level of satisfaction – and his willingness to entertain such a judgment as feasible. When the point is reached at which *all* group members feel this way as a minimal criterion you may assume that you have reached a consensus as it is defined here and the judgment may be entered as a group decision. This means, in effect, that a single person can block the group if he thinks it necessary; at the same time, it is assumed that this option will be employed in the best sense of reciprocity. Here are some guidelines to use in achieving consensus:
>
> 1. Avoid *arguing* for your own rankings. Present your position as lucidly and logically as possible, but consider seriously the reactions of the group in any subsequent presentations of the same point.
> 2. Avoid *'win-lose'* stalemates in the discussion of rankings. Discard the notion that someone must win and someone must lose in the discussion; when impasses occur, look for the next most acceptable alternative for both parties.
> 3. Avoid changing your mind *only* in order to avoid conflict and to reach agreement and harmony. Withstand pressures to yield which have no objective or logically sound foundation. Strive for enlightened flexibility; avoid outright capitulation.
> 4. Avoid conflict-reducing techniques such as the majority vote, averaging, bargaining, coin flipping, and the like. Treat differences of opinion as indicative of an incomplete sharing of relevant information on someone's part and press for additional sharing, either about task or emotional data, where it seems in order.
> 5. View differences of opinion as both natural and helpful rather than as a hindrance in decision making. Generally, the more ideas expressed the greater the likelihood of conflict will be; but the richer the array of resources will be as well.
> 6. View initial agreement as suspect. Explore the reasons underlying apparent agreements; make sure that people have arived at similar solutions for either the same basic reasons or for complementary reasons before incorporating such solutions in the group decision.

These instructions were designed, of course, to operationalize the normative system thought to underlie the behaviors observed by Hall & Williams (1970) in their study of trained groups. While it was expected that various skill-lapses would limit the effectiveness with which subjects would be able to implement their instructions, the major focus of the instructions was upon the attitudes of members toward group potential and the legitimacy of certain group behaviors which are required for the constructive resolution of conflicts. To the extent that the instructions were effective in promoting different orientations toward group action on the part of experimental groups which, in turn, would result in the employment of different practices, performance differences should be discernable between control and experimental groups.

CRITERIA OF MEASUREMENT

The NASA Moon Survival Problem affords a number of estimates of group functioning. Not only can the adequacy of group decisions be evaluated, but indices of utilization of resources, creativity, and achievement of the synergy effect bonus may be computed as well.

Decision adequacy index. Since subjects' decisions amount to rank orderings of standard items, it is possible to compare individual and group orders with the expert rank ordering supplied by NASA as a means of quantifying decision adequacy. The adequacy of a decision can be assessed in terms of its summed deviations from the expert rank order; this summed deviation, of course, represents an error score, the magnitude of which is inversely related to decision quality. Thus, the task affords a strictly numerical index of decision adequacy, at both the individual and group levels, which is free to vary from 0 to 112 points away from absolute accuracy. Normative studies indicate that the average individual error score obtained is 39.30 with a standard deviation of 6.62 points.

Utilization of resources index. The rank order data of individual members – both in terms of the unique ordering which characterizes each person's decision and the error score which results – may be taken as an index of a group's prediscussion resources. The pooled error scores reflect the average quality of available resources prior to discussion, and the ranks given to a particular item per se by all members yield an item-by-item index of the resources available for decision making on each. By comparing the array of individual judgments on an item which was available before group discussion with the group judgment on that item, it is possible to infer something about whose judgments were finally incorporated by the group, the quality of these, and the proportion of the group typically holding a particular judgment before its incorporation. This content analysis approach reveals a general pattern of resource utilization for the groups in both conditions.

Creativity index. When groups abandon their prediscussion resources in favor of emergent insights and solutions which are suggested during interaction, opportunities for creativity may result. The incorporation by groups of judgments not held by any member prior to discussion has been found to vary with the amount of substantive conflict present (Hall & Williams, 1966), and may reflect creative processes to the extent that the emergent product moves the group toward a qualitatively better decision than it would have had had it continued to rely on existing resources. Thus, the incidence of incorporation of emergent judgments alone tells very little about a group's creativity; rather, it is the qualitative significance of the product which determines whether creativity may be inferred. This index may be based on both frequency measures – *i.e.*, the number of times emergent solutions were qualitatively superior to existing resources – and qualitative difference scores obtained from a comparison of emergent error scores with those representative of existing resources.

Synergism index. Closely associated with group creativity is the achievement of synergy bonus. As productivity which surpasses both the sum of individual efforts and that of a group's most proficient member, this bonus has been coordinated to the effective integration of task and social-emotional issues by Collins & Guetzkow (1964). To the extent that a group decision product is qualitatively superior to that of its most accurate member's, credit may be given for group synergy. The index may be expressed for both group conditions in either frequency terms or as quality increments derived from a comparison of group and most proficient member error scores.

Summarily, four criteria of measurement are available for comparing the performances of control and experimental groups in the present study.

RESULTS

The general hypothesis considered was that groups which had received a normative intervention (instructed groups) would perform more effectively on a decision-making task than groups which had received no intervention (uninstructed groups). The normative intervention employed was predicted to legitimize certain behaviors on the part of group members which, although they are thought to be essential for creative and productive group action, are typically viewed as deviant and disruptive acts within groups. At the same time, the intervention was designed to promote the use of more constructive techniques for resolving substantive conflicts and to block or discourage reliance on more traditional methods which actually serve to suppress opinion differences before they can be fully explored. It was felt that instructed groups, to the extent that they were able to adhere to the normative instructions given them, would engage in more searching behaviors, converge less quickly on decision alternatives, participate more uniformly, experience greater concern about the opinions and feelings of individual members, and generally gear their contributions to a higher definition of group potential than would the more tradition-bound uninstructed groups.

As a test of the degree to which groups in both conditions functioned as expected, all groups were polled immediately subsequent to completion of the task about the manner in which they had been able to work. Of the 16 instructed groups, 13 indicated that they had adhered closely to the process outlined in their instructions; three groups reported considerable difficulty in following the instructions and reported resorting to a majority vote as they became concerned about time pressures. Of the 16 uninstructed control groups, 15 reported almost total reliance upon a majority vote procedure, with the size of the majority varying from one decision issue to the next. The one remaining group described what was essentially a consensus approach not unlike that employed by the instructed groups. In addition, uninstructed groups reported that they were not aware that other groups had received instructions or special attention of any sort. In general, therefore, it would seem that instructed and uninstructed groups employed processes which differed in enough respects that the normative intervention may be assumed to have been successful in altering group functioning, and a test of the effects of the intervention on decision quality, utilization of resources, creativity, and achievement of the synergy effect bonus may be made.

THE EFFECT OF NORMATIVE INSTRUCTION ON GROUP DECISION QUALITY

The first hypothesis stated that instructed groups would produce decisions which were qualitatively superior to those produced by uninstructed groups. Implicit in this hypothesis was the assumption that the prediscussion resources available to instructed and uninstructed groups in fashioning their decisions would be of comparable quality. Inspection of the average resource data obtained from the groups, however, revealed that the pre-discussion resources of instructed groups were slightly superior to those of uninstructed groups, despite the care taken in equating groups during their composition. Although the difference between the mean prediscussion resources of the groups (error scores of 45.07 and 47.52 for instructed and uninstructed groups, respectively) was not found to be significant, the fact that there was any prediscussion advantage on the part of instructed groups suggested that the hypothesis could be tested most objectively by the use of covariance procedures.

The prediscussion data, taken as a base line of substantive competence for the groups, and the final group decision scores were subjected to an analysis of covariance. A preliminary analysis revealed that the regression lines for instructed and uninstructed groups could be considered to be generally parallel within the limits of random sampling, and a final analysis for treatment effects was performed. The mean error score obtained by instructed groups was 25.94, while the mean error score obtained by uninstructed groups was 34.19 points. This mean difference in decision quality of 8.25 points yielded an F of 4.20 (1/30 df) which is significant at the .05 level. Thus, it may be concluded that instructed groups produce significantly superior decisions on the NASA Moon Survival Task as compared with uninstructed groups after adjustment for the quality of prediscussion resources. The data, therefore, may be interpreted as supporting Hypothesis 1.

UTILIZATION OF AVAILABLE RESOURCES UNDER DIFFERENT NORMATIVE CONDITIONS

Much of the superiority in decision quality expected from instructed groups was predicated on the assumption that these groups would make more effective use of their available resources. As Collins & Guetzkow (1964) have pointed out, one of the conditions inherent in group decision making which serves to enhance the quality of group decisions is the availability of a relatively wider range of information and opinions which, in turn, increases the probability that group members will hit upon more adequate solutions. Implicit in this observation is the assumption that groups will be able to recognize their more adequate resources when they are expressed and, moreover, that the probability of expression is high. Several studies have shown, however, that the probability of expression of opinions is often a function of their commonality. Thus, groups which employ interaction practices which facilitate the expression of ideas irrespective of their commonality should be expected to make even better use of available resources than groups operating under the commonality constraint. Since the normative intervention employed with experimental groups was expected to promote a freer expression of ideas and opinions, it was predicted in Hypothesis 2 that instructed groups would utilize their available resources more effectively than uninstructed groups would, simply because of the increased probability that these resources would be expressed and recognized.

Since the individual prediscussion judgments of all group members were collected prior to group decision making in the administration of the NASA task, it was possible to determine how resources were utilized by groups by comparing the prediscussion member data with the content of the final group decision. Given the 15 interdependent judgments required of subjects in reaching a decision, utilization of resources can be assessed in several ways. For example, it is to be expected that for many of the judgments to be made groups will have available among their members at least one person with the objectively correct judgment. The frequency with which groups actually incorporate this resource in their final decision may serve as one index of utilization of available resources. By the same token, when the objectively correct judgment is not held by the membership it is to be expected that one or more of the members will possess nearly correct judgments; the frequency with which such 'best available' judgments are used by groups may constitute a second-order index of resource utilization. Finally, a more gross indicant of the manner in which available resources have been employed by groups may be obtained from the error score which results when the data are combined from those group judgments reflecting reliance on available member resources. As tests of Hypothesis 2, analyses were made of each of these three measures.

Utilization of objectively correct resources. Of the 15 judgments which groups were required to make in reaching a decision, both instructed and uninstructed groups had the correct judgment represented among their member resources 48 per cent of the time. Inspection of the data revealed that instructed and uninstructed groups employed the correct judgment with exactly the same frequency. Both groups selected the correct judgment 41 per cent of the time when it was available, or an average of 2.95 times out of a possible 7.20 times. The lack of difference between groups on this measure would tend to discredit Hypothesis 2 unless differences in the use of 'best available' resources are found to exist.

Utilization of 'best available' resources. Since an objectively correct resource was available in fewer than half the decision instances comprising the group task, a more practical test of Hypothesis 2 may be the manner in which groups used the 'best' resource available when a correct judgment was unavailable. The prediscussion data of members were again analyzed relative to the final group decision in order to ascertain to what extent groups would continue to rely on available resources and, moreover, to select the 'best available' judgments from these in the absence of correct resources. The frequencies with which the two group types continued to rely on existing resources in making a judgment were again found to be identical; both groups incorporated judgments from existing member data in 48 per cent of the cases – or an average of 3.74 out of 7.80 times – when no correct resource was available. Instructed groups were found to use the best of these existing resources an average of 1.76 times as compared with the mean frequency of 1.31 times for uninstructed groups. A chi square analysis of these data resulted in a nonsignificant value (χ^2 = 1.24) and Hypothesis 2 was further discredited.

Error scores from judgments based on available resources. As a final test of the second hypothesis (and as a means of further verifying the results obtained from the analyses of frequency data just presented), decision error scores were computed for all groups on the basis of those final group judgments which reflected the use of preexistent member resources. This procedure yielded a mean decision error score for available resource-based judgments of 14.4 for instructed groups and a mean score of 15.4 for uninstructed groups. The *F* obtained from a comparison of these mean scores was less than 1.00. Thus, Hypothesis 2 fails to receive support, and it must be concluded that the normative intervention employed does not affect the manner in which groups utilize their available resources in reaching decisions, at least with respect to performance on the NASA Moon Survival Task.

Summarily, it may be said that with respect to their use of available resources there are no differences between instructed and uninstructed groups which will account for the decision-making superiority noted on behalf of the instructed groups in the present study. This suggests that the significant performance increment achieved by instructed groups may be found to stem from those decision instances in which groups abandoned existing resources in favor of novel and emergent judgments which they created for themselves. The creation of emergent judgments is the focus of Hypothesis 3.

CREATIVITY IN INSTRUCTED AND UNINSTRUCTED GROUPS

When the correct judgment is not available to groups, they are left with two options: they may continue to rely on existing resources in an attempt to select the best available judgment or they may create new judgments for use in the final decision. It has already been observed that the choice of the former option does not differentiate between instructed and uninstructed groups; by implication, neither does the choice of the latter option differentiate between groups in terms of the sheer frequency with which new judgments were created. Both group types resorted to the use of emergent judgments some 27 per cent of the time in reaching decisions. The significance of employing emergent judgments does not lie in frequency counts, however; rather, it is the extent to which the creation of emergent judgments is a reflection of the type of 'aha' experience which is likely to occur during group interaction that lends significance to the use of emergent judgments. Abandoning existing opinions and judgments may free groups to hit upon elegant solutions or at least to improve over the quality of existing resources. Creativity of this sort is essentially a case of 'increasing likelihoods,' since on the NASA task – as on most decision tasks – there is no way for groups to know with certainty that they either do or do not possess the necessary resources for producing qualitatively adequate decisions. Thus, the assumption is that a tolerance for divergent opinions may lead to heightened and more flexible interaction among group members which, in turn, will increase the likelihood that novel and qualitatively superior insights will be stimulated. Since one aim of the normative intervention employed with instructed groups was to promote (if not require) increased tolerance for divergent thinking, it was predicted in Hypothesis 3 that instructed groups would achieve greater creativity than uninstructed groups would. Although it was not intended that the hypothesis should bear such a crucial burden, it now appears that the difference in creativity predicted under Hypothesis 3 may well be the discriminating feature in the performance of instructed versus uninstructed groups. Because of this possibility, Hypothesis 3 was tested in several different ways.

Creativity and the achievement of elegant solutions. The first test of Hypothesis 3 was suggested by the negative results obtained in testing the second hypothesis. Those instances in which correct resources were available combined with the instances in which groups relied on existing member data accounted for 73 per cent of the decision-making activities of both instructed and uninstructed groups. For the remaining 27 per cent of the decision, groups created solutions. To the extent that instructed groups created qualitatively better solutions than uninstructed groups did, some insight into the performance differential characterizing the groups may be gained.

One index of the quality of groups' emergent judgments — and probably the most direct measure — is the frequency with which such judgments resulted in the achievement of an elegant solution or, put differently, the selection of the objectively correct judgment despite its zero probability of occurence. Instructed groups were found to incorporate emergent judgments which were objectively correct some 36 per cent of the time. By contrast, uninstructed groups created correct judgments only 19 per cent of the time. A comparison of these mean frequencies, with the data corrected for discontinuity, yielded an χ^2 of 3.85 (1 df) which is significant at the .05 level. Thus, instructed groups, although they do not employ emergent judgments any more frequently, would seem to make more constructive use of their creative products than uninstructed groups do. In terms of the achievement of elegant solutions, the data may be interpreted as supporting Hypothesis 3.

Quality of emergent judgments relative to average available resources. Aside from resulting in the achievement of elegant solutions, it is possible that the use of emergent judgments may simply result in a group incorporating judgments which are generally superior to the average resource available. The manner in which groups employ their emergent judgments — whether for purposes of compromise or creativity — has been suggested by Hall & Williams (1966) to be a determinant of the extent to which such judgments will constitute an improvement or gain in accuracy over the quality of available resources. Since instructed groups were expected to avoid mechanical compromises and to seek creative solutions, it was hypothesized that their emergent products would be significantly superior to those of uninstructed groups in terms of the gain in accuracy over average resources represented by these judgments.

Instructed groups produced emergent judgments which averaged 5.9 points more accurate than the average available resource would have been had it been used on the solutions in question. Uninstructed groups, on the other hand, incorporated emergent judgments in their decisions which averaged 1.5 points better than the average resource available. A comparison of these mean scores yielded an F of 4.734 (1 & 30 df) which is significant at the .05 level. Thus, the emergent gain score data are in the direction predicted under Hypothesis 3, and it may be inferred that instructed groups experienced greater creativity than did uninstructed groups, as creativity is reflected by the gain score measure.

A related question was also explored with respect to the emergent gains created by instructed and uninstructed groups; given an improvement over average resources on the part of both groups, a question arises as to whether emergent gains are in fact significant improvements over available resources. While instructed groups experienced significantly greater gains than uninstructed groups did, it is possible that neither type group significantly improved its decision product as a result of its use of emergent judgments. To test this possibility, correlated t tests were made of the differences in accuracy between average resource data and emergent products for groups under the two normative conditions of interest. The mean improvement of 5.9 points experienced by instructed groups produced a direct difference t of 3.922 (15 df) which, with a one tailed test, is significant at the .005 level. By comparison, the mean difference score between average resource and emergent judgment quality of 1.5 points obtained by uninstructed groups failed to yield a significant value ($t < 1.00$). Thus, the inference drawn is that instructed groups benefit significantly from their use of emergent judgments while uninstructed groups do not realize any appreciable increment over their average resources from using emergent judgments. These results would seem to lend further support to Hypothesis 3.

Summarily, it appears that instructed groups realize greater returns from those judgments which they create than uninstructed groups do. To the extent that a more frequent achievement of elegant solutions and significant improvements over available resources through the use of emergent judgments may be taken as indices of group creativity, the data confirm the prediction of greater creativity on the part of instructed groups. In retrospect – with particular reference to the disconfirmation of Hypothesis 2 – it would seem that creativity via emergent judgment is the most discriminating factor in comparing the performances of instructed and uninstructed groups. That such creativity could be promoted by a relatively simple normative intervention would seem to lend additional credence to McGregor's (1967) comments on the creation of effective groups. A final performance consequence of such an intervention is the subject of Hypothesis 4.

ACHIEVEMENT OF THE SYNERGY BONUS AS A FUNCTION OF NORMATIVE INSTRUCTIONS

Closely associated with group creativity of the type covered under the third hypothesis is achievement of group synergy. Collins & Guetzkow (1964) have suggested that when groups effectively integrate the task and interpersonal concerns inherent in a decision-making situation they are likely to experience increments in decision quality above and beyond that which would be expected on the basis of either the sum of individual efforts or the performance of a group's most skilled member. This synergy bonus is seen as potentially available to all groups, and its achievement is limited only by the manner in which group members work together. Since the normative instructions employed in the study were designed to guide the members of instructed groups toward more cooperative practices and to facilitate the integration of task and emotional concerns, it was predicted in Hypothesis 4 that instructed groups would achieve the synergy bonus more frequently than uninstructed groups would.

Of the 16 instructed groups, 12 or 75 percent surpassed the performance of their most proficient members in fashioning a decision on the NASA task. By comparison, of the 16 uninstructed groups, only 4 or 25 per cent were able to produce decisions which were superior to their most skilled members'. A comparison of these frequencies, with the data corrected for discontinuity, yielded an χ^2 of 6.125 ($p < .02$, 1 df). From this value it may be concluded that the difference between instructed and uninstructed groups in achievement of group synergy is a result of the difference in normative treatments, and Hypothesis 4 is confirmed.

While the surpassing of a group's most proficient member in fashioning a decision is certainly an important index of group effectiveness, analyses based on the frequency with which this effect was achieved tell very little about the qualitative increments implied in the group versus most accurate individual comparison. The data were further analyzed to determine the qualitative significance of achieving (or failing to achieve) the synergy bonus. The group decision error scores were compared with those of the most proficient members for both instructed and uninstructed groups in order to ascertain whether achievement of group synergy was necessarily associated with a *significant* gain over a group's most accurate resource.

The most proficient resources of instructed and uninstructed groups were comparable; mean error scores of 31.13 and 31.69 were obtained by the most accurate persons in instructed and uninstructed groups, respectively. Instructed groups, on the average, were found to produce decisions which were 5.09 points more accurate than the decisions held by their most accurate group members. A direct difference t test of this mean value produced a t of 2.544 which, with 15 df, is significant beyond the .025 level. Uninstructed groups, on the other hand, produced decisions which averaged 2.50 points less accurate than those of their most proficient members; an analysis of this difference score yielded a t of $-.97$, which is not significant. Thus, the inference drawn is that instructed groups improve significantly over the performance of their superior resources in producing decisions while uninstructed groups, although they fail to utilize all of their resources, produce decisions which do not differ significantly from those of their most accurate members.

Summarily, it would seem from the present results that it is possible to modify a group's orientation to its decision task by introducing a number of specific normative guidelines. Moreover, it appears that a modification of this type can affect the manner in which groups will actually work to such an extent that they will perform more effectively than groups with unmodified, more intuitive, approaches to the task. Obviously, the degree to which this effect will be achieved is a direct function of the theoretical validity of the intervention design and the feasibility of the strategy employed. Since rationale and strategy constitute critical factors in any intervention attempt, some attention should be given to the assumptions and inferences underlying the design of the intervention employed in the present study.

DISCUSSION

Three of the four major hypotheses tested were supported by the data; in each of these, predictions favoring the performance of instructed over uninstructed groups were confirmed. Instructed groups were found to produce qualitatively better decisions, to be more creative, and to achieve the synergy bonus more frequently and by a greater margin than were uninstructed groups. These differences were apparently due to the normative intervention employed with instructed groups. Just why a relatively simple – and certainly unenforceable – statement of procedural groundrules should elicit performance differences of the type observed cannot be explained without a good deal of speculation about the predispositions of members of uninstructed groups and the net effect of normative models aimed at the systemic elements of group process. While speculation may risk oversimplification, the inferred dynamics of uninstructed groups can be discussed as a basis for designing interventions, and the intended effects of the present normative model can be identified and evaluated in the light of the obtained results.

It has already been pointed out that the normative intervention employed was designed to break what appeared to be an implicitly shared strain toward convergence on the part of group members and to guide groups through the resolution of opinion conflicts which were likely to emerge when such a strain was interrupted. Implicit in this view are two assumptions: first, it was assumed that individuals place inordinate value on convergence because they equate it with group movement, efficiency, and harmony; secondly, it was inferred from the observation of a number of naive groups that individuals tend to equate conflict and opinion differences with personal rejection. In a general sense it was felt that individual members – while often confronted with group situations – are ambivalent with respect to handling the social-emotional and task obstacles which emanate from different approaches to decision making. Essentially, therefore, the normative intervention employed was designed to modify the response biases of individual members of decision-making groups.

Benne & Sheats (1948) have suggested that people approach group membership with certain attitudes toward group work, more or less conscious skills for dealing with others, and fairly well developed rationales of group process. These are essentially issues of response bias; and, while the origin of such attitudes, skill preferences, and rationales is a matter worthy of research in its own right, there are enough similarities in functioning as one moves from group to group to suggest the presence of some system of shared values and implicitly accepted groundrules which insure group movement while affording a modicum of protection for members. This normative system – if in fact it does exist – may be construed as a major determinant of member response biases. Given the preoccupation with democratic processes which characterizes individuals in this culture, it might be inferred that the normative system which prevails in groups – or which individuals are most predisposed to invoke – is one of majority rule. Moreover, it is likely that this technique will be especially used in resolving opinion differences. Thus, it was anticipated that uninstructed groups would gravitate naturally toward the use of majority pressures at the first sign of a disruption of group mobility. The fact that 15 of the 16 uninstructed groups reported a reliance on the majority vote technique of decision making would seem to substantiate this expectation.

Not only are there quality issues inherent in the majority rule approach to decision making, but there are social-emotional and attitudinal issues as well. Barnlund (1959) has reported that the majority technique yields decisions which are inferior in quality to those produced under a consensual technique, and Daval (1967) has recently reported that a review of historical events suggests that majority decisions approximate in quality those produced by a group's average member. Thus, under the normative system toward which group members are thought to be most inclined, there is a forfeiture of resources and creativity. Forfeiture of any sort is likely to affect the attitudes of group members toward group work; these, in turn, affect the goals which members set for their groups – that is, what they deem as possible – and the behaviors which they feel are 'appropriate' for themselves and others. In general, the goals seem to be of a lower level than is realistic and the behaviors sanctioned are rather non-confronting, self-abrogating, and uncreative; the result, it was reasoned, is that the typical group is composed of individuals who are inclined to accept moderately good decision products fashioned by moderately committed people.

Thus, breaking the strain toward convergence was but a first-step objective in the modification of member response biases. The major intent in the design of the normative intervention was one of overcoming member prophecies of moderate productivity before they could be fulfilled. By blocking the use of more traditional techniques and by initiating a number of practices which, if followed, would increase the probability that members would pursue their differences until genuine agreement could be reached, the intervention was intended simply to legitimize the use of those behaviors deemed necessary for more complete resolution of differences – regardless of how confronting or self-centered they might appear. Implicit sanctions were re-directed toward what were previously social imperatives, and members became obliged to press for solutions that were personally acceptable. To the extent that individuals were able to withstand pressures to yield, it was felt that this would prevent closure and force groups to remain open to alternatives. In retrospect, it seems logical that creativity was the distinguishing feature between groups functioning under instructed and uninstructed conditions.

Summarily, the present study would seem to suggest that group performance is very much a function of the particular conditions under which groups work; these conditions, in turn, are controlled to a great extent by the inclinations and procedural orientations of the collective membership. Finally, although the results were generally in the expected direction, qualifications are in order due to the nature of the design. The intervention employed reflected a systems approach to modifying a group system. That is, the normative instructions were composed of a cluster of suggestions which, although they were related systematically, may have had either the multiple impact intended or only isolated impacts traceable to one or two of the instructions. Given the significant improvement of instructed over uninstructed group performance, additional research seems warranted in an attempt to tie down the specific elements – if there are such – of the normative instructions which are responsible for the performance increment noted for instructed groups. Similarly, despite the attempt to equate conditions for the instructed and uninstructed groups in general, the demand characteristics of the normative intervention may have been such that the instructed groups simply tried harder than the uninstructed groups did. While the fact that both group types took the same amount of time to complete the task would suggest that this was not the case, it remains a possibility in need of study. Therefore, the major implication of the present investigation seems to be that it is possible to upgrade the performance of a relatively naive group through the use of a planned intervention aimed at the systemic elements of group process. Issues of what constitutes a best intervention, the significance of demand characteristics, single versus clustered normative suggestions, and the like may represent the research questions raised by the present study.

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BIOGRAPHICAL NOTES

JAY HALL received his doctorate in social psychology from the University of Texas in 1963. He was an Associate Professor in the Graduate School of Business when the present research was conducted, and is presently Director of the American Behavioral Science Training Laboratories and President of Teleometrics, Inc., both in Houston, Texas.

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APPENDIX

DECISION FORM AND KEY[2]

NAME ________________

GROUP ______________

Instructions: You are a member of a space crew originally scheduled to rendezvous with a mother ship on the lighted surface of the moon. Due to mechanical difficulties, however, your ship was forced to land at a spot some two hundred miles from the rendezvous point. During the crash landing, much of the equipment aboard was damaged and, since survival depends on reaching the mother ship, the most critical items available must be chosen for the two hundred mile trip. Below are listed the 15 items left intact and undamaged after landing. Your task is to rank order them in terms of their importance in allowing your crew to reach the rendezvous point. Place the number *1* by the most important item, the number *2* by the second most important and so on through number *15*, the least important.

15	Box of matches	*Useless since there is no oxygen on the moon.*
4	Food concentrate	*Satisfies basic energy requirements*
6	50 feet of nylon rope	*Useful in scaling cliffs, tying injured together, etc.*
8	Parachute silk	*Protection from sun's rays*
13	Portable heating unit	*Only useful if on the dark side of the moon.*
11	Two .45 calibre pistols	*Possible source of self-propulsion*
12	1 case dehydrated Pet milk	*Duplicates food concentrate in bulkier form*
1	2 hundred-pound tanks of oxygen	*Absolute necessity for life support*
3	Stellar map (of the moon's constellation)	*Most important means of determining position and directions*
9	Life raft	*CO_2 bottle possible propulsion device*

2. Numbers shown give key to the correct order of importance.

14	Magnetic Compass	*Virtually useless since magnetic field on the moon isn't polarized*
2	5 gallons of water	*Absolute necessity to sustain life*
10	Signal flares	*Possible distress signal once close enough to mother ship to be seen*
7	First aid kit containing injection needles	*Injection needles fitted to suit aperture quite useful*
5	Solar-powered FM receiver-transmitter	*Only useful if line-of-sight transmission is possible with limited transmission range*